Global Data Center Resource Usage

Water & Electricity Consumption | EDA Summary Report

KEY METRICS

PUE	1.643	1.057	2.000
Energy Metric	Global Median PUE	Most Efficient	Least Efficient

WHAT WAS ANALYSED

- Dataset loaded; shape, info, and statistical summary checked via describe()
- PUE global distribution plotted — histogram with KDE reveals two distinct efficiency tiers
- WUE global distribution plotted — wide spread reflects variable cooling strategies globally
- PUE and WUE compared across Facility Types using bar plots and box plots
- Year-on-Year PUE and WUE improvement tracked via violin plots (2019–2025)
- Water Stress Tier density tracked year-on-year with a stacked bar chart
- The top 15 countries by data center count and the top 10 by total capacity (MW) were identified
- Cooling system type distribution analysed via countplot across all facilities
- PUE vs. Daily Electricity Usage and Capacity vs. Daily Water Usage correlations plotted

KEY FINDINGS

!	**High Global Median PUE**	Median PUE of 1.643 means over half the world's data centers waste significant energy on cooling and power overhead, not computing.
!	**PUE 2.0 — Critical Waste**	The maximum recorded PUE is 2.0 — these facilities use as much power on overhead as on actual IT workloads, with serious cost and carbon implications.
~	**Two Distinct Efficiency Tiers**	PUE clusters at ~1.2 (hyperscale) and 1.5–1.6 (enterprise), revealing a systemic technology and investment gap across the global industry.
~	**Cooling Drives Water Consumption**	WUE varies significantly by cooling type. Evaporative cooling is far more water-intensive than air-side economizers — rarely optimised at the planning stage.
~	**Expansion Into Water-Stressed Areas**	Year-on-year data shows acceleration of data center growth in high and extremely high water stress regions, outpacing sustainability frameworks.
+	**Linear Capacity–Water Relationship**	Capacity (MW) and daily water usage show a near-linear correlation — larger facilities must plan water sourcing at the same scale as power infrastructure.

RECOMMENDATIONS FOR IMPROVEMENT

1	**Mandate PUE Targets Below 1.4**	Enforce PUE benchmarks for all new builds. Hyperscale standards of ~1.2 should become the industry norm, not the exception.
2	**Shift to Air-Side Economizers**	Facilities in temperate climates should prioritise air-side economisers over evaporative cooling to cut WUE dramatically without performance loss.
3	**Restrict Siting in Water-Stressed Zones**	Siting policies should incorporate water stress tier data. High-stress locations must require water recycling and alternative sourcing plans before approval.
4	**Retrofit Legacy Enterprise Facilities**	The 1.5–1.6 PUE cluster represents thousands of facilities. Modernising with liquid immersion or hot-aisle containment can reduce energy waste by 20–30%.
5	**Build a Predictive Resource Model**	Train a regression model on facility type, location, cooling system, and capacity to predict PUE and WUE for smarter infrastructure planning.

Analysis performed using Python (Numpy, Pandas, Matplotlib, Seaborn) | Dataset: data_center_hybrid.csv